

Text, OCR, or Pixels?

A Retrieval Decision Matrix for Quant-Finance Documents, Built on Eighteen Years of the osQF Archive

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Abstract

Quantitative-finance knowledge lives disproportionately in visually dense documents—broker research, filings, and conference slide decks full of tables, charts, and equations that defeat plain text extraction. We use a public, domain-matched testbed to measure what actually works: the complete osQF/R-Finance archive, 626 talks spanning 2009–2025. We build a fully local pipeline—resumable R stages for scraping and extraction, a 14B open-weight LLM performing forced tool-call extraction with verbatim evidence-quote validation, and a DuckDB semantic index—and then quantify three retrieval layers against each other with a vision-judged evaluation: native text extraction, VLM-OCR re-extraction (a judged bake-off between two 2026 open-weight OCR models, with an acceptance gate that guarantees no regressions), and pixel-level page-image retrieval that answers chart questions both OCR models score 0.00 on. The result is a practical decision matrix—text, OCR, or pixels, by document class—that transfers directly to practitioner document stores. Everything runs on one consumer GPU with zero cloud calls. We release the pipeline, the index, and a hosted MCP endpoint so attendees can query eighteen years of this conference’s own archive from their AI tooling, live, during the talk.

Keywords: retrieval-augmented generation, document AI, OCR, local LLMs, DuckDB, open source, quantitative finance

1 Introduction

The archived editions of this conference—R/Finance 2009–2019 and osQF 2022–2025—have produced hundreds of slide decks and papers, publicly hosted at <https://www.osqf.org/archive/>, and effectively unsearchable. Finding “that talk where someone benchmarked GARCH estimation libraries” means paging through year indexes and opening PDFs one at a time. The archive is a knowledge base without a query interface.

This is not a librarian’s problem; it is the same problem every quantitative desk has. Broker research, vendor decks, prospectuses, and filings share the osQF archive’s exact pathologies: born-digital text next to image-rendered text, dense tables, charts whose content exists only in pixels, formulas, code snippets, and two decades of drifting file formats (including legacy binary .ppt that modern parsers refuse). Retrieval systems that assume clean text silently lose exactly the content practitioners care about most—the numbers in the exhibit.

We therefore treat the conference’s own archive as a domain-matched, legally clean, fully public testbed, and ask a practical question: *for each kind of content, which retrieval representation actually works—extracted text, VLM-OCR output, or the raw page image?* Our contributions:

1. An open, reproducible pipeline that turns the complete osQF archive (626 talks, 2009–2025) into a structured, semantically searchable knowledge base with *verbatim evidence-quote validation*, running entirely on one consumer GPU (§??).
2. A quantified, vision-judged comparison of three retrieval layers—text, VLM-OCR (including a bake-off of two 2026 open-weight OCR models), and pixel-level page-image retrieval (§??).
3. A decision matrix mapping document classes to retrieval layers, with the evidence behind each cell (§??).
4. Associative (multi-hop) retrieval over the archive’s talk–topic–speaker graph, implemented natively in R (§??).
5. Open artifacts: MIT-licensed code, the downloadable index, and a hosted read-only MCP endpoint so any attendee’s AI tooling can query the archive during the talk (§??).

2 Corpus and Pipeline

The osQF archive pages for 2009–2025 (no conference was held 2020–2021) yield 835 agenda rows, which reduce to **626 real talks** after filtering event headers (breaks, lunches, panels). 468 talks have retrievable files: 413 PDF, 65 PPTX, 8 legacy binary PPT, 1 PPTM, and 27 HTML pages; 25 files are corrupt at the source (dead links from the pre-2020 hosting). The pipeline (Figure ??) is five resumable R stages; every stage is idempotent and cache-backed, so partial failures resume rather than restart.

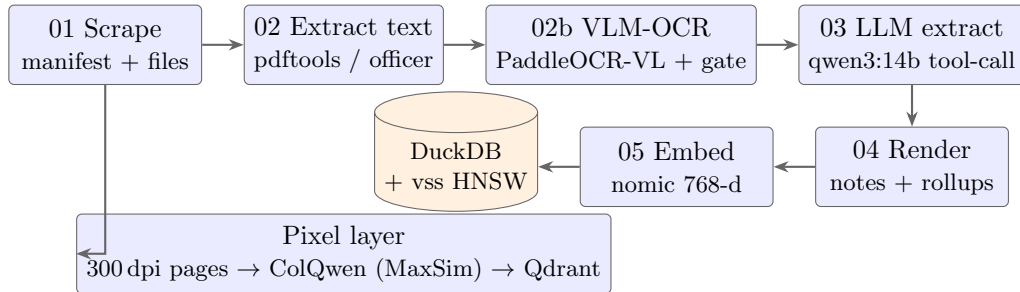


Figure 1: The five-stage text pipeline plus the parallel pixel layer. A single 300-dpi render of all 462 extractable decks (12,276 page images) feeds both the OCR repair stage and pixel retrieval.

Structured extraction with evidence quotes. Stage 03 sends each talk’s text to a locally served 14B open-weight model (qwen3:14b via Ollama) with a *forced tool-call schema*: headline, topics, methods, datasets, code/tools referenced, key findings, applications, references. The critical design rule is **do not infer**: every key finding must carry an `evidence_quote` that is a verbatim substring of the source text (≤ 200 chars), validated post-hoc by whitespace-normalized substring matching. The corpus-wide evidence pass rate is **69.2%**, and—because it is computed mechanically—it doubles as a per-talk trust label on every rendered note. Failed validations are visible, not silent.

Semantic index. Rendered notes are chunked by section and embedded with nomic-embed-text (768-d) into DuckDB with the `vss` extension (HNSW, cosine): **4,111 chunks** across 462 talks, queryable with metadata filters (year, speaker, topic, method) in a single SQL expression. DuckDB is the deliberate choice over a vector-database service: the entire index is one 24 MB file that ships as a release asset.

3 Three Retrieval Layers

3.1 Layer 1: Native text

`pdftools` and `officer` handle born-digital documents well and are the cheapest layer by orders of magnitude. They fail in three specific ways: image-rendered text and SmartArt in slide decks (thin extractions), garbled text layers in old PDFs, and legacy `.ppt` (unreadable). Before repair, 25 talks had failed extraction outright and dozens more were thin or garbled.

3.2 Layer 2: VLM-OCR, with a judged bake-off

We converted all 74 PowerPoint decks to PDF (LibreOffice headless, closing the legacy-`.ppt` gap), rendered every extractable deck to 300-dpi page images, and re-extracted the corpus with an open-weight vision-language OCR model served by vLLM on the same GPU. Two 2026 open-weight candidates were bake-tested on 15 decks stratified across the known failure classes (garbled PDFs, legacy PPT, thin PPTX, zero-evidence extractions), with a frontier vision model judging 37 dense pages against the rendered page image (Table ??).

	Unlimited-OCR (3B MoE)	PaddleOCR-VL (0.9B)
Fidelity, all judged pages	0.48	0.76
Fidelity, chart pages	0.54	0.57
Pages with hallucinated content	13/37	4/37
Downstream evidence-quote pass	39.7%	70.1%
Model size (BF16)	6.4 GB	2.0 GB

Table 1: Vision-judged OCR bake-off on 15 stratified osQF decks. The larger, newer model loses on every criterion that matters downstream; both models are near-useless on charts (§??). A practical finding: both models respond only to their trained task prompts (“OCR:”, “Free OCR.”)—free-form instructions return empty output.

After the initial bake-off we added the current OmniDocBench leader (MinerU2.5, layout-routed hybrid parsing [?]) as a third arm on the identical pages: fidelity tied the champion (0.74 vs. 0.76), hallucinations tied (3–3), and downstream evidence-quote pass trailed by 4.6pp—the champion held. Leaderboard rank did not predict our-corpus rank, which is the argument for judging candidates on your own documents.

Two engineering guards matter more than the model choice. First, an *acceptance gate*: OCR output replaces the native text layer only if it is ≥ 300 characters *and* $\geq 0.7\times$ the native extraction’s length—349 of 462 decks accepted, 113 rejected (mostly 2009–2011 L^AT_EX beamer decks whose native text layer is better than any OCR). Second, a *best-of-both revert*: if downstream structured extraction fails on the OCR text where it previously succeeded, the deck reverts to its pre-OCR state. The re-extraction is therefore monotone: it recovered 19 previously dead talks (including all legacy `.ppt`) and regressed none. A caveat this design makes explicit: for OCR-sourced notes, evidence quotes validate against OCR output rather than source truth, so hallucinated OCR could silently “validate”—which is why the bake-off’s hallucination criterion is weighted above raw fidelity.

3.3 Layer 3: Pixel retrieval

Charts defeat both OCR models (fidelity ≈ 0.55 , and frequently 0.00 on pure-plot pages). The pixel layer skips transcription entirely: all **12,276 page images** are embedded directly into a

Qdrant collection; text queries embed into the same space, and a small local VLM (Qwen2.5-VL-3B) reads the retrieved page image to produce an answer. Retrieval began as single-vector page embeddings (Qwen3-VL-Embedding-2B, 2048-d) and was subsequently migrated to *late-interaction* multi-vector retrieval (ColQwen2.5: MaxSim over sequence-pooled patch embeddings, 128-d per vector), following the ColPali design [?]. In a judged-relevance evaluation (12 queries; a frontier vision model grading each retrieved page), late interaction lifts precision@1 from 0.75 to **0.83** and precision@3 from 0.67 to **0.75**, with the gain concentrated entirely in fine-grained visual-detail queries: “model-selection tables with AIC/BIC columns” goes from 0/3 relevant pages retrieved to 3/3, exactly the regime the late-interaction literature predicts. In our earlier controlled A/B on research-note PDFs, pixel retrieval beat text retrieval by **+0.50** accuracy on table questions while tying on prose. On this corpus the effect is starker: for a chart question about a 2011 talk’s oil-price/housing scatter plot—a page where *both* OCR models scored 0.00 fidelity—pixel retrieval returns the correct pages at ranks 1–3, and the local reader correctly describes the axes, ranges, and the quadratic fit. The content was never in any text layer; it was only ever in the pixels.

4 The Decision Matrix

Content / document class	Layer	Evidence
Born-digital text (filings, papers, beamer decks)	Native text	113/462 decks: native beats OCR; zero cost
Image-rendered text, SmartArt, scans, legacy formats	VLM-OCR + acceptance gate	349/462 accepted; 19 dead talks recovered; gate blocks regressions
Tables (broker grids, exhibits)	OCR if gated; pixels when values must be exact	OCR-VL 0.76 fidelity; pixel +0.50 vs. text on table Q&A
Charts and plots	Pixels only	both OCR models ≈ 0.55 overall, 0.00 on plot pages; pixel retrieves rank-1
Formulas	VLM-OCR (L ^A T _E X output)	supported by both models; validate downstream
Cross-document “who/what relates to X”	Graph + PPR (§??)	expert-finding validated on 462 talks

Table 2: Which representation to retrieve from, by content class. The mapping transfers directly from conference decks to practitioner stores: broker research is the PPTX column, filings are the born-digital column, vendor chart books are the pixel column.

The general principle: *extraction is lossy in proportion to visual density, and the loss is silent*. The engineering answer is not one perfect extractor but a layered system in which each document is served by the cheapest layer that preserves its content—with mechanical gates (length ratios, evidence-quote validation, judged fidelity) deciding, per document, when to escalate.

5 Associative Retrieval: the Archive as a Graph

Semantic search answers “find talks about *X*.” It cannot answer “who, across eighteen years, is the person to ask about *X*?”—the association routes through entities, not text similarity. We build

a talk–entity graph (462 talks; 985 nodes; 890 edges over shared topics, methods, and speakers) and run Personalized PageRank from cosine-seeded talks, natively in R via `igraph`. Three design choices, found the hard way, that a sparse curated graph needs: entity-only edges (dense kNN edges collapse PPR into degree centrality), IDF-weighted edges ($w = \log N/\text{deg}$, so one shared rare topic outweighs a hub speaker), and ranking by PPR *lift* over the unpersonalized baseline (de-emphasizing celebrities). Queries like “robust portfolio optimization” then surface the archive’s actual specialists—the researchers who return to the topic across editions—rather than the most prolific speakers.

6 Hardware and Cost

Every stage—scraping, extraction, the OCR bake-off and full re-extraction, embedding, judging setup, and serving—ran on a single RTX 3090 Ti (24 GB) under WSL2, with open-weight models and zero cloud calls in the data path.¹ Approximate wall-clock: full-corpus render 35 minutes (CPU); full-corpus OCR ~ 4 hours; structured re-extraction ~ 2.5 hours; pixel embedding ~ 3 hours; the semantic index rebuilds incrementally in minutes. The GPU is the only scarce resource, and stages are sequenced so no two model servers contend for it. A desk could run the identical stack against its own document store without any data leaving the building—which, for broker research and internal documents, is not an optimization but a requirement.

7 Reproducibility and Access

All artifacts accompany this paper:

- **Code** (MIT): the full pipeline, evaluation scripts, and MCP server at <https://github.com/travisjakel/osqf-archive>.
- **Index**: a single self-contained DuckDB release asset (~ 50 MB) holding the semantic index, talk manifest, derived notes, tag tables, precomputed expert/related-talk graphs, and a persisted full-text index—usable offline with zero model downloads via the built-in BM25 fallback, or with full semantic search after one `ollama pull`.
- **Hosted MCP endpoint, live now**: a read-only Model Context Protocol server over the complete archive (semantic search, talk notes, expert-finding, guarded SQL) at <https://osqf.assetflow.ai>, so attendees can point Claude, or any MCP client, at eighteen years of this conference during the talk. The shared attendee token appears on the talk’s closing slide; a reviewer token accompanies this submission. Client setup: `osqf_mcp/CLIENT_SETUP.md` in the repository.
- **Provenance**: original speaker PDFs are *not* redistributed; notes carry short validated quotes and link to osqf.org sources. Model weights are pulled from their upstream repositories; licenses are catalogued in `NOTICE.md`.

¹The only cloud use in the project is the bake-off *judge* (a frontier vision model grading OCR transcripts against page images), acceptable here because the corpus is public conference material. The equivalent local judge (Qwen2.5-VL) is included in the release for private corpora.

8 Related Work and Conclusion

Pixel-level document retrieval follows the direction of ColPali [?]; our associative layer is a native-R adaptation of HippoRAG’s PPR-over-entity-graph design [?]; the OCR candidates are PaddleOCR-VL [?] and Unlimited-OCR [?]. What this paper adds is not a new model but a *measured decision procedure* for combining these layers over a real, messy, eighteen-year document store—with gates that make the system monotone (repairs never regress) and honest (evidence quotes are mechanically validated, and their limits stated).

The osQF archive was the testbed, but the deliverable is the pattern: layered retrieval with per-document escalation, judged on your own documents rather than leaderboards, running entirely on hardware you control. The conference’s collective memory is now queryable—by everyone in the room.

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